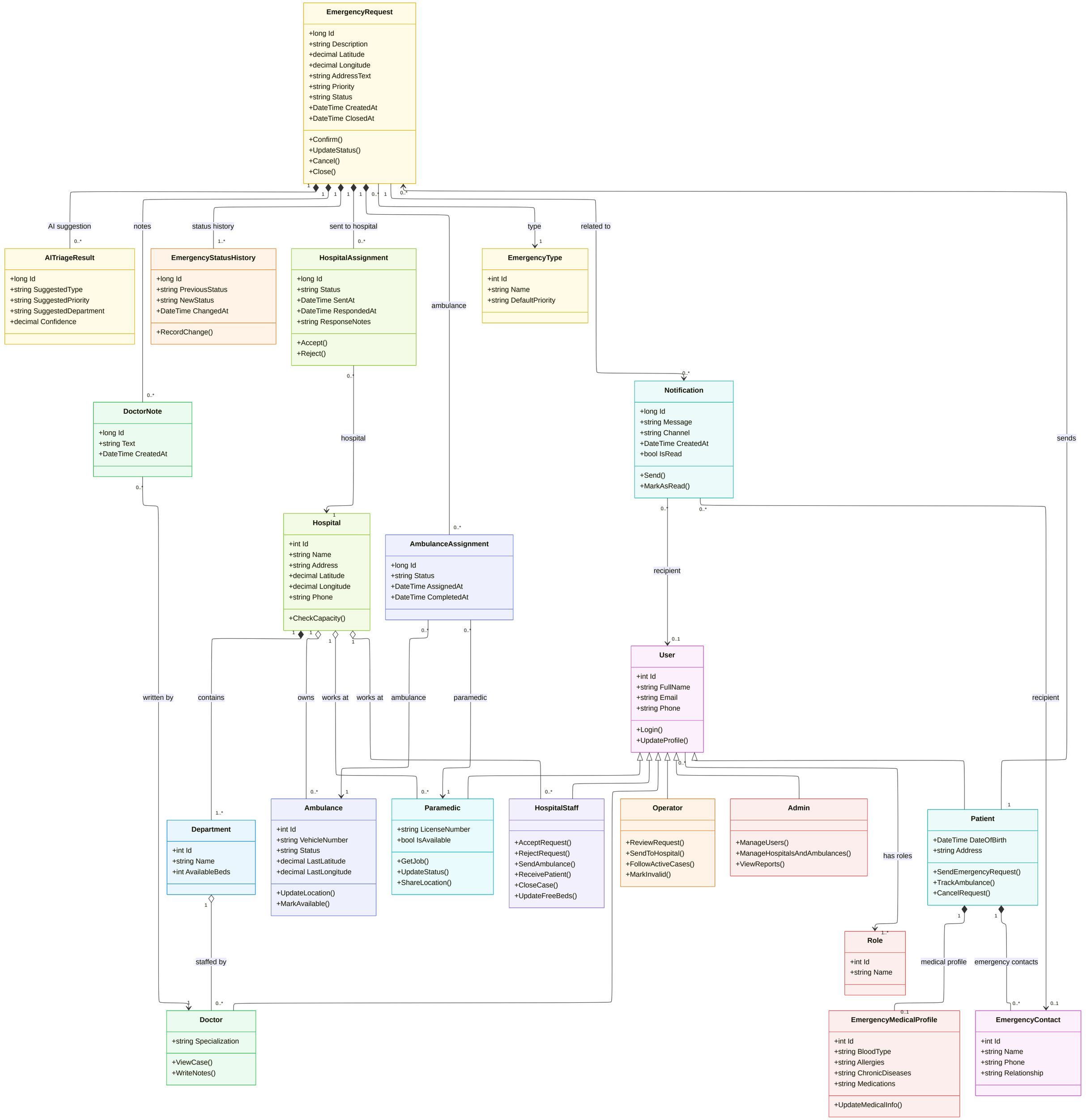
# Class Diagram

Smart Emergency Response & Healthcare Coordination Platform



## Main Classes

- **User:** common account info (name, email, phone). It is the parent class of all the roles.
- **Role:** the roles a user has, used for the permissions.
- **Patient:** sends the emergency request, has a medical profile and emergency contacts, and tracks the ambulance.
- **Operator:** reviews the request and the AI suggestion, then sends the request to a hospital.
- **HospitalStaff:** accepts or rejects the request, sends the hospital's ambulance with a paramedic, receives the patient, closes the case and updates the free beds.
- **Paramedic:** gets the job from his hospital, updates the case status and shares his location.
- **Doctor:** views the case and writes his notes.
- **Admin:** manages users, hospitals and ambulances, and views reports.
- **EmergencyRequest:** the emergency case. It has the description, the location (GPS or the address the patient wrote), the priority and the status.
- **EmergencyType:** the list of emergency types (accident, chest pain, ...).
- **AITriageResult:** the AI suggestion for the type, priority and department. It is only a suggestion, the operator decides.
- **HospitalAssignment:** every time the request is sent to a hospital. Its status is Pending, Accepted or Rejected, so if a hospital rejects, we keep it and send to the next closest one.
- **Hospital:** a hospital with its location.
- **Department:** a hospital unit and its free beds.
- **Ambulance:** a vehicle that belongs to a hospital, with its status and last location.
- **AmbulanceAssignment:** which ambulance and which paramedic went to the case.
- **EmergencyMedicalProfile:** blood type, allergies, chronic diseases and medications.
- **EmergencyContact:** the patient's family members who get messages.
- **DoctorNote:** the doctor's notes on the case.
- **Notification:** a message sent to a user or to an emergency contact.
- **EmergencyStatusHistory:** every status change of the request.

## Relationships

- User is the parent class of Patient, Operator, HospitalStaff, Paramedic, Doctor and Admin.
- A user can have one or more roles.
- A patient has one optional medical profile and many emergency contacts.
- A patient can send many emergency requests.
- Each request has one emergency type and can have AI suggestions.
- A request can be sent to many hospitals (one at a time) until one accepts.
- A hospital contains departments, and doctors work in a department.
- A hospital owns its ambulances and has its own paramedics and hospital staff.
- After the hospital accepts, an ambulance and a paramedic are assigned to the request.
- Each request keeps its status history, the doctor's notes and its notifications.
- A notification goes to a user or to an emergency contact.

## Request Statuses

Submitted, Confirmed, Sent to Hospital, Ambulance Assigned, On the Way, Arrived, Picked Up, Going to Hospital, At Hospital, Completed. A request can also be Cancelled (before an ambulance is assigned) or Invalid (marked by the operator).